

Heavy-Tailed Initialisation as a New Paradigm in Continual Learning

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Abstract

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Statements and Acknowledgements

Statement of Originality

I certify that this thesis contains work carried out by myself except where otherwise acknowledged.

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Acknowledgements

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Statement of Contribution

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Introduction

Deep learning has achieved unprecedented successes across a vast array of modern domains, from powering conversational language models to accelerating medical diagnostics and autonomous robotics. However, the paradigm underpinning these breakthroughs relies almost exclusively on static training regimes over fixed, independent, and identically distributed (i.i.d.) datasets. True intelligent systems must instead possess the capacity for continual learning—the ability to sequentially acquire, consolidate, and adapt knowledge over time from non-stationary data streams without retraining from scratch. This lifelong learning capability is essential for real-world deployments where environments evolve dynamically, such as personal robotics, real-time medical monitoring, and adaptive edge computing, while serving as a fundamental prerequisite for mimicking biological intelligence. Yet, transitioning neural networks from static optimization to continual learning uncovers a profound vulnerability: when adapted to new tasks, models systematically overwrite previously acquired representations, encountering the fundamental obstacle known as the stability-plasticity dilemma.

The stability-plasticity dilemma reflects a trade-off in how connectionist architectures encode and retain information. To learn a new task, a neural network requires sufficient plasticity to update its weight parameters and integrate novel feature representations. Conversely, to retain past knowledge, the network demands stability to preserve the specific parametric configurations and decision boundaries established during prior training. In standard gradient-based optimization, parameter updates driven by a new task loss vector act destructively on the weight spaces dedicated to earlier tasks, precipitating an abrupt collapse in past performance known as catastrophic forgetting. Because standard deep architectures treat parameter space as a shared, unconstrained substrate, improving plasticity almost invariably degrades stability, forcing current models onto an unfavorable Pareto frontier between learning capacity and memory retention. To navigate this trade-off, researchers over the past decade have devised a broad spectrum of continual learning strategies designed to enforce memory preservation.

While these algorithmic interventions have yielded significant empirical progress, they share a fundamental design choice: they act strictly as post-initialisation interventions. Across these diverse paradigms, the initial state of the network is treated as a passive housekeeping detail, relying almost universally on standard isotropic Gaussian schemes such as Xavier or Kaiming initialisation to prevent unstable gradients at step zero. However, this reliance on Gaussian initialisations assumes that isotropic weight distributions are the optimal starting substrate for non-stationary optimization—an assumption that remains largely unexamined. By treating initialisation as a static prerequisite rather than an active structural hyperparameter, existing methods attempt to patch representation collapse without considering whether the underlying initial weight geometry itself accelerates catastrophic forgetting. Indeed, recent theoretical developments suggest that highly performant, well-regularized neural networks naturally diverge from these conventional Gaussian assumptions.

This divergence is precisely illuminated by the framework of Heavy-Tailed Self-Regularization (HTSR) introduced by Martin and Mahoney. Their empirical and theoretical analysis demonstrates that as deep networks train and successfully generalize, their Empirical Spectral Densities (ESDs) universally evolve away from random-matrix-like bulk distributions toward heavy-tailed power-law decay—a regime indicating intrinsic self-regularization without explicit parameter penalties. Complementing these spectral signatures, recent work has shown that the weight matrices of state-of-the-art architectures are best modeled by α -stable distributions, while the optimization dynamics of Stochastic Gradient Descent (SGD) themselves exhibit non-Gaussian, heavy-tailed noise characteristics known as Lévy flights. These discrete, heavy-tailed jumps enable the optimizer to traverse high-dimensional loss surfaces more efficiently, escaping sharp,

suboptimal local minima in favour of wider, more resilient basins. If heavy-tailed representations and dynamics represent the universal, self-regularized end-state of well-trained models, initiating training from isotropic Gaussian distributions forces the network to spend significant optimization capacity overcoming an artificial initial geometry. Consequently, configuring the initial weight substrate using heavy-tailed distributions—specifically symmetric α -stable initialisations—emerges as a compelling, fundamental paradigm to structurally pre-condition neural networks for non-stationary, continual learning environments.

In this thesis...

Chapter 1: Literature Review

1.1 Deep Learning Fundamentals

1.1.1 Neural Networks

In essence, a neural network is a computational model inspired by the structure and functions of biological neural networks in the brain. They consist of many units called *neurons*, each one connected to other neurons by *edges*. Each neuron receives signals from connected neurons, then processes them and sends a signal to other connected neurons. The signal is a real number; specifically, a neuron computes a weighted sum of its inputs plus a *bias* term, which is then passed through a non-linear *activation function* to determine the output. The strength of the signal at each connection is determined by a *weight*, which adjusts during the training process. Normally, neurons are organised into layers. Signals travel from the first layer (*the input layer*) to the last layer (*the output layer*), passing through intermediate layers (*hidden layers*) along the way. A network is typically called a deep neural network if it has multiple hidden layers. This is illustrated below in Figure 1.1.

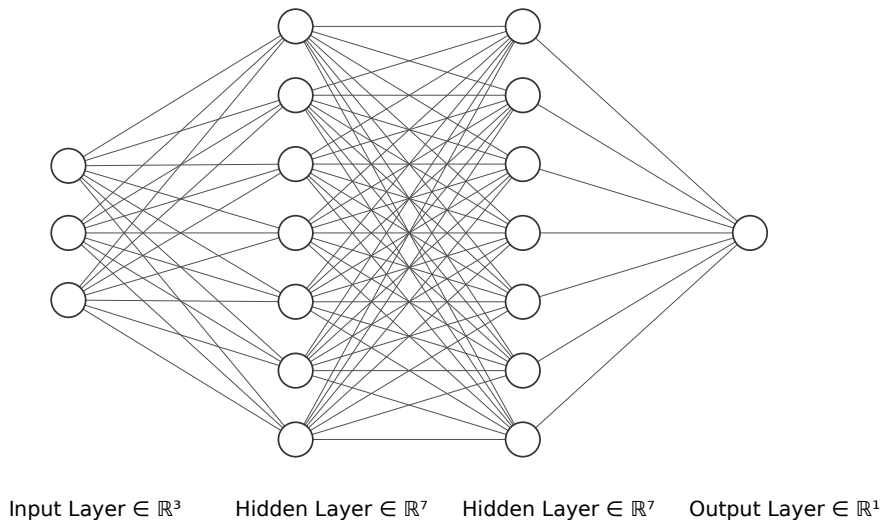


Figure 1.1: A schematic of a deep neural network with an input layer, two hidden layers, and an output layer, depicting the neurons and edges.

The fundamental power of neural networks is built on the *Universal Approximation Theorem* (UAT), which states that neural networks with a certain structure can, in principle, approximate any continuous function to any desired degree of accuracy. This theorem provides a mathematical justification for using DNNs, as a sufficiently large or deep network can model any abstract relationship found in real-world data. The UAT is however, a non-constructive existence theorem, which only guarantees that such a network is possible, not how feasible it is. It does not provide a method for finding the network's parameters, nor does it specify exactly how large the network must be for a given function. Finding a suitable network remains a practical challenge that is typically addressed with various optimisation algorithms like backpropagation and gradient descent. Consequently, the study of deep learning shifts from a question of representational existence to one of dynamical navigation: how a network's state evolves through a high-dimensional parameter space to find these elusive solutions.

1.1.2 Training and Evaluation

The Goal of Training

The fundamental goal of training a neural network is improve its mapping of the input to the desired output, by adjusting the weights of the model. This can be understood as the network being an approximator of a function that maps an input x_i to the output y_i . We denote the function represented by the network as F , where $z_i = F(x_i; \mathbf{w})$ is the prediction corresponding to the input x_i for a given set of weights $\mathbf{w}$ that parameterises the network. The process of training is effectively described as minimising the error between the set of predictions $\{z_i \mid i = 1, 2, \dots, N\}$ and the set of true values $\{y_i \mid i = 1, 2, \dots, N\}$. To quantify this error, a *loss function* is used,

$$L(\mathbf{w}) = \frac{1}{N} \sum_{i=1}^N d(y_i, z_i) \quad (1.1)$$

where $d(y_i, z_i)$ is a representation of the error between prediction z_i and true value y_i . Various types of loss functions are used in the field of deep learning, from Mean Squared Error used for regression tasks, to Cross-Entropy Loss for classification tasks. In any case, any well performing network should minimise this value, and the goal of training is adjusting the parameters $\mathbf{w}$ to accomplish this.

The Learning Process

In practice, a model is trained by learning from an example set of representative data, called the *training set*. We denote this as $\{x_i, y_i \mid i = 1, 2, \dots, N\}$, where x_i is the data point and y_i is the ground truth label. This data is then sent through the network in what is known as a *forward pass*. The output of the model z_i is then used as an input into Eq. 1.1 alongside the associated y_i , returning a loss. Now, the core question is: how do we adjust the weights $\mathbf{w}$ to decrease the loss? This is accomplished by the process of *backpropagation*, an efficient application of the multivariate chain rule to neural networks. It efficiently computes the gradient of the loss with respect to the network weights, by propagating derivatives backward (*backward pass*), one layer at a time. By assigning a gradient to each weight within a network, which points in the direction of higher loss, we can move in opposite direction to decrease our loss. In principle, this can be accomplished through an algorithm called gradient descent (GD) that prescribes for the evolution of network weights,

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla L(\mathbf{w}_t) \quad (1.2)$$

where $\mathbf{w}_t$ are the weights at step t , ∇L is the gradient of the loss and η is the *learning rate*, a *hyperparameter* (a parameter set before training) governing step sizes. In practice however, gradient descent is computationally expensive, requiring the forward pass of the whole training set during each iteration, and prone to poor performance. Therefore, an alternative algorithm to GD called stochastic gradient descent (SGD) is used instead:

$$\mathbf{w}_{t+1} = \mathbf{w}_t - \eta \nabla \tilde{L}_t(\mathbf{w}_t) \quad (1.3)$$

Here, the stochastic gradient $\nabla \tilde{L}_t$ is an estimate of the true gradient ∇L calculated using a *mini-batch*. The mini-batch is a random subset $\Omega_t \subset \{x_i, y_i \mid i = 1, 2, \dots, N\}$ of the training set with the hyperparameter *batch size* S . Training is effectively an iterative loop of choosing a mini-batch, doing a forward pass, calculating the loss, doing backpropagation, and moving the weights opposite to the gradient. From a physical perspective, this stochasticity introduces a thermal-like fluctuation into the weight evolution, effectively turning the optimiser into a particle undergoing a random walk driven by the local landscape geometry. The training of a network is

typically terminated once the performance converges to a sufficient value, or a set number of *epochs*, where one epoch is defined to be $\lceil \frac{N}{S} \rceil$ iterations.

Evaluation and Generalisation

However, a crucial flaw in the training process is *overfitting*, where the network adapts so well to the training set that it starts to struggle on previously unseen data, making it unsuitable for real-world use. Therefore, a separate pool of data points is reserved as a *test set*, denoted as $\{\chi_i, v_i \mid 1, 2, \dots, M\}$. The test set does not contribute to the evolution of the weights during training. Instead, it is used to evaluate generalisation, how well the network performs on previously unseen data. This makes it the primary diagnostic to prevent overfitting. The generalisation is measured by the testing accuracy, which is defined as the percentage of predictions $F(\chi_i; \mathbf{w})$ that match the true value v_i for a set of weights $\mathbf{w}$. In almost all applications of neural networks, testing accuracy is valued more than training accuracy or loss. This preference for generalisation is often linked to the "flatness" of the local minima reached by the optimiser, a property we will explore through the lens of multi-fractal landscape structure in later sections.

1.1.3 The Geometry of Learning

The Loss Landscape

A popular and powerful method to understand the learning process of a neural network, is to view it as a particle navigating through a complex, high-dimensional *loss landscape*. The loss landscape $L(\mathbf{w})$ is a mathematical surface mapping a neural network's parameters to its loss value. Every possible permutation of the weights $\mathbf{w}$ maps to a point on the surface, and the height of that point represents the loss. Therefore in principle, the training process thus involves finding the global minimum (the lowest point) of the loss. In practice however, for deep neural networks, the loss landscape is a highly non-convex, high-dimensional surface. It is characterised by chaotic terrain containing numerous local minima, sharp valleys, and wide, flat regions. A *local minima* is a point on the loss landscape where the loss is lower than its immediate neighbours, meaning the gradient is zero. This leads to stalled learning even when performance is poor. Of particular importance are *saddle points*, points on the landscape where the gradients in orthogonal directions are all zero, but which is not a local extremum of the function. They are a primary driver of complex learning dynamics and phase transitions in deep learning. Overall, the learning process is highly non-trivial, and in practice, finding a sufficient local minima is the goal of optimisation.

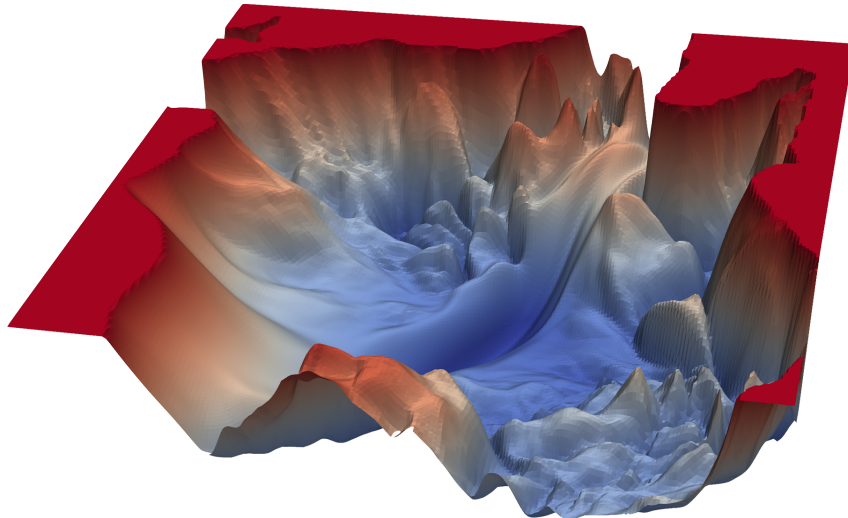


Figure 1.2: A visualisation of the loss landscape for VGG-56. Regions of high and low loss are shaded in red and blue respectively. Note the rugged, non-convex nature of the landscape.

Hessians and Jacobians

A powerful mathematical tool used to study and characterise the loss landscape is the *Hessian matrix* $\mathbf{H} = \nabla^2 L(\mathbf{w})$, defined as the square matrix of second-order partial derivatives of a scalar-valued function, in this case the loss function. It describes the local curvature of the loss landscape, and is used to characterise many features like local minima and saddle points. Specifically, the eigenvalues λ_i of the Hessian reveal the "steepness" of the landscape in the vicinity of a critical point. The maximum eigenvalue λ_{max} is a common proxy for the *sharpness* of a minimum, while the full singular value spectrum provides insight into how the network is compressing information. The *Flat Minima Conjecture* posits that flatter regions with lower curvature and smaller maximum eigenvalues are strongly correlated with superior generalisation and stability. In the context of this thesis, we track these spectral observables to monitor the structural re-organisation of weights during the learning process.

While the Hessian describes the overall curvature of the loss, the *Jacobian matrix* $\mathbf{J}^l = \frac{\partial \mathbf{x}^l}{\partial \mathbf{x}^{l-1}}$ characterises the internal propagation of signals between layers. The total sensitivity of the network's output to its input is determined by the product of these layer-wise operators. In conventional Gaussian networks, these products can lead to the vanishing or exploding gradient problem if not carefully fine-tuned. Therefore, we utilise the layer-wise Jacobian as a primary diagnostic tool to monitor the internal "health" of signal propagation across the network's depth, looking for structural signatures that might indicate a balance between expansion and contraction.

Optimisation Heuristics and the Generalisation Gap

To navigate these high-dimensional landscapes efficiently, various heuristics and regularisation techniques are typically employed. *L2 regularisation*, or weight decay, adds a penalty term to the loss function to prevent the weights $\mathbf{w}$ from growing excessively, effectively smoothing the landscape to prevent overfitting. Furthermore, while *vanilla SGD* uses a fixed learning rate η , adaptive optimisers such as *Adam* adjust the step size for each parameter individually based on the history of previous gradients.

In the experimental setup of this thesis, we purposefully omit these advanced heuristics, utilising vanilla SGD with no weight decay. This minimalist protocol ensures that any observed rich learning behaviour or structural evolution is an intrinsic result of the heavy-tailed initialisation rather than an artifact of the optimisation algorithm. The success of the model is ultimately measured by the *generalisation gap*: the difference between the model's performance on the training set versus the *test set* of previously unseen data. A smaller gap indicates that the network has truly learned abstract features rather than simply memorising the training noise.

1.2 Dynamical and Spectral Theories of Deep Learning

1.2.1 Mean Field Theory

Mean Field Theory (MFT) is a core mathematical tool from physics and probability theory, used to study the behaviour of high-dimensional stochastic models. The core concept of MFT is to replace all interactions to any one body with an average or effective interaction, sometimes called a *mean field*. By reducing a complex system to a simpler model that approximates the original via the mean field, it reduces any many-body problem into an effective one-body problem. The increased ease of study means that insight into the behaviour of the system can be obtained at a lower computational cost.

This is highly powerful for the field of deep learning, where even the simplest models have millions of interacting parameters. We treat the weights as independent and identically distributed (i.e.i.d.) Gaussian variables. This allows us to invoke the *Central Limit Theorem* (CLT), thereby approximating the behaviour of any layer as a *Gaussian distribution*. A Gaussian can be fully

described by just its mean μ and variance σ^2 , greatly simplifying the many neurons and edges within a layer. In practice, with most initialisation schemes setting $\mu = 0$ to avoid shifting the signal, we are primarily concerned with the variance σ^2 . By studying how σ^2 varies across a network, we can predict the growth or shrinkage of a signal as it passes through the model.

This is formalised via the *iterative map*, which describes the variance of the *pre-activations* at the *Mean Field Limit* ($N \rightarrow \infty$):

$$q^l = \sigma_w^2 \int_{-\infty}^{\infty} \frac{dz}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} \phi\left(\sqrt{q^{l-1}}z\right)^2 + \sigma_b^2 \quad (1.4)$$

In Eq. 1.4, q^l denotes the variance (or second moment) of the neural inputs at layer l . The parameters σ_w^2 and σ_b^2 represent the weight and bias variances, respectively, which dictate the coupling strength between layers. In the mean field limit ($N \rightarrow \infty$), this map converges to a deterministic recursion relation that characterises the fluctuations of the neural input distribution throughout the depth of the network.

The primary utility of this iterative approach is the identification of the fixed point q^* , where $q^l = q^{l-1}$ for large l . By linearising the network around this stationary state, one can derive the input-output Jacobian, which serves as a diagnostic for signal stability. This framework allows researchers to determine whether a network will suffer from the vanishing or exploding gradient problem before training ever begins, providing a principled guide for hyperparameter selection.

1.2.2 Deep Information Propagation

The study of how signals evolve through the layers of a deep network is known as *Deep Information Propagation*. Using the MFT framework, the dynamical behaviour of a DNN can be categorised into three distinct phases based on the weight variance σ_w^2 and the resulting spectral radius of the layer-wise Jacobian.

1. **The Ordered Phase:** Also referred to as the silent regime, this occurs when σ_w^2 is low. In this phase, signal correlations increase and the magnitude of activations exponentially decays (vanishes) as they propagate deeper into the network. Geometrically, this corresponds to a uniform contraction of the internal neural representations.
2. **The Chaotic Phase:** This occurs when σ_w^2 exceeds a critical threshold. Here, small perturbations in the input are exponentially amplified (explode), leading to a chaotic expansion of representations where the signal eventually becomes indistinguishable from noise.
3. **The Edge of Chaos:** This is the critical transition line between the ordered and chaotic regimes. At this point, information can propagate to theoretically infinite depths without vanishing or exploding. This phase is characterised by a balance of contraction and expansion, enabling the network to learn complex, abstract mappings.

Quantitatively, for Gaussian random networks, the transition to chaos is identified by the maximum singular value (spectral radius) of the layer-wise Jacobian crossing unity. Specifically, in the Gaussian limit ($\alpha = 2$), the edge-of-chaos transition is predicted to occur at a single point or thin line in the parameter space, such as $D_w = 1$ for networks with zero bias.

However, this classical Gaussian framework possesses a significant limitation: the "fine-tuning problem". Because the critical regime in Gaussian models is restricted to a thin boundary, any slight deviation in parameters can push the network into a non-trainable state. Furthermore, Gaussian initialisations result in delocalised eigenvectors that spread evenly across neural sites, lacking the mixture of localised and delocalised properties seen in more complex biological systems. This suggests that while Gaussian MFT provides a clean baseline, it may be insufficient for describing the robust, self-organising criticality observed in real-world deep learning architectures.

1.2.3 The Neural Tangent Kernel and Learning Regimes

While Mean Field Theory provides a static snapshot of signal propagation, the *Neural Tangent Kernel* (NTK) framework describes the continuous-time evolution of the network during training. In the infinite-width limit ($N \rightarrow \infty$), the dynamics of a deep neural network undergo a remarkable simplification: the weights move only infinitesimally from their initial values, yet the network’s output evolves according to a linear model. This regime, often termed *lazy learning*, is exceptionally powerful as it implies that the complex, non-linear training trajectory of a DNN can be described by a static, constant kernel. Consequently, this allows for the use of kernel methods and closed-form algebra to predict training outcomes and generalisation performance without the computational requirement of executing gradient descent.

The core of this framework is the NTK matrix, $\Theta^{(L)}$, which is defined by the inner product of the network gradients with respect to the weights:

$$\Theta^{(L)}(\mathbf{x}, \mathbf{x}') = \sum_{p=1}^P \frac{\partial F(\mathbf{x}; \mathbf{w})}{\partial w_p} \frac{\partial F(\mathbf{x}'; \mathbf{w})}{\partial w_p} \quad (1.5)$$

In Eq. 1.5, F represents the network function and P is the total number of parameters. Under the assumption of infinite width, this kernel remains frozen at its initial value throughout the entire training process. This leads to a simple linear differential equation governing the evolution of the network’s predictions over time t :

$$\frac{dF(\mathbf{x}, t)}{dt} = -\eta \Theta^{(L)}(\mathbf{x}, \mathbf{x}_j) \nabla_F L \quad (1.6)$$

Eq. 1.6 suggests that the learning process is equivalent to a kernel regression where the features are fixed at the start of training. This provides a precise signal-to-noise threshold for identifying learning boundaries. However, the NTK limit represents a “lazy” regime where the network fails to perform true *feature learning*, as the underlying representations do not adapt to the structure of the data.

In contrast, most state-of-the-art deep networks operate in the *rich learning* (or feature learning) regime. In this state, the weights undergo significant structural re-organisation, allowing the network to actively learn and specialise its internal representations. Characterising the transition between these two regimes is a central challenge in modern theory. This study utilises these frameworks to define learning boundaries and identify the mechanisms, such as spatial decoupling, that allow different layers to exhibit either lazy or rich learning characteristics during the training trajectory.

1.2.4 The Edge of Stability

The interaction between the learning rate η and the local geometry of the loss landscape defines the stability of the optimisation trajectory. Central to this interaction is the concept of *sharpness*, a metric derived from the Hessian matrix $\mathbf{H}$. Quantitatively, sharpness is defined by the maximum eigenvalue of the Hessian, λ_{max} , which represents the direction of steepest curvature in the high-dimensional parameter space. Low values of λ_{max} correspond to flatter, more resilient regions of the landscape, while high values indicate sharp, narrow valleys where the gradient changes rapidly over small distances.

Classical optimization theory, largely built upon the study of quadratic and convex objectives, prescribes a strict upper bound on the learning rate to ensure convergence. For a function with local smoothness λ_{max} , Gradient Descent is expected to converge only when the learning rate satisfies $\eta < 2/\lambda_{max}$. In this classical regime, exceeding this threshold is predicted to result in

a catastrophic divergence of the loss, as the optimizer overshoots the minimum and enters an unstable, exploding trajectory.

However, empirical observations in deep learning have revealed a phenomenon that defies this classical prescription, known as the *Edge of Stability* (EoS). During the training of non-convex deep networks, it is observed that the landscape undergoes "progressive sharpening," where λ_{max} increases until it reaches the critical threshold of $2/\eta$. At this boundary, the network does not diverge; instead, λ_{max} begins to hover just above the $2/\eta$ limit. In this EoS regime, the system exhibits local instability, characterised by short-term oscillations and increases in loss, yet manages to maintain long-term convergence and self-stabilisation.

This dynamical behaviour is hypothesised to be a primary mechanism by which neural networks navigate toward flatter minima. By operating at the edge of stability, the optimiser effectively "bounces" out of sharper, sub-optimal regions until it encounters a basin wide enough to accommodate the learning rate without local divergence. In the context of this thesis, we monitor the evolution of spectral observables, such as the spectral norm and the stable rank, to identify how these stability thresholds interact with the non-equilibrium dynamics induced by heavy-tailed initialisation.

1.2.5 Transition to Multifractality

The classical frameworks of Gaussian Mean Field Theory and convex optimisation provide a mathematically convenient baseline, yet they appear increasingly insufficient to describe the reality of deep learning dynamics. Empirical observations of phenomena such as the *Edge of Stability*, "grokking", and particularly the aging *anomalous diffusion* of SGD suggests that the loss landscape is far more complex than the smooth, well-behaved surfaces assumed by established theory. Furthermore, the discovery of chaotic dynamics in deterministic gradient descent implies that stochastic-like behaviour can emerge purely from interactions with a multi-scale landscape structure. These anomalies indicate that the origins of generalisation may be fundamentally rooted in the geometric structure of the loss landscape itself.

To model this heterogeneous complexity, we look to the field of fractal geometry. A *fractal* is a mathematical object that exhibits scale-invariance, meaning its structure remains similar when viewed at different magnifications. While monofractals are characterised by a single, global scaling exponent, real-world disordered media often require an extended spectrum of exponents, a property known as *multifractality*. In the context of a loss landscape, multifractality implies that the roughness of the surface varies heterogeneously across the parameter space.

Quantitatively, this roughness can be described using the *pointwise Hölder exponent*, $H(\mathbf{x})$, which serves as a local measure of regularity. For a multifractional Brownian surface (MBS), the local fractal dimension D is related to this exponent via $D = d + 1 - H(\mathbf{x})$, where d is the topological dimension. Landscapes with low $H(\mathbf{x})$ values are characterised by high roughness and anti-persistent increments, whereas high $H(\mathbf{x})$ values represent smoother regions with persistent correlations.

The current lack of a theory that accounts for these empirical observations while explaining generalisation remains a significant gap in the literature. Traditional Langevin models are contingent upon external stochasticity and fail to predict the time-dependent, non-equilibrium dynamics observed in actual networks. There is a clear demand for a new, landscape-focused theoretical framework that can reconcile these complex dynamical signatures. By shifting the focus from Gaussian statistics to heavy-tailed distributions and multifractal geometries, we can begin to uncover the mechanisms that enable the robust computational performance of deep architectures.

1.3 Lévy Mean Field Theory

While Gaussian Mean Field Theory provides a rigorous baseline for signal propagation, its reliance on the Central Limit Theorem (CLT) restricts its applicability to networks with finite-variance weight distributions. However, empirical evidence suggests that deep networks in practice often exhibit heavy-tailed coupling weights that are better described by α -stable distributions. To account for these statistics, recent work has developed *Lévy Mean Field Theory*, which generalises the Gaussian framework to the regime of infinite variance ($\alpha < 2$).

1.3.1 Stable Distributions and the Scale Parameter

In this framework, the weight entries W_{ij}^l are sampled from a symmetric α -stable distribution $S_\alpha(\sigma)$, where $\alpha \in (1, 2]$ dictates the thickness of the tails. In the Gaussian limit ($\alpha = 2$), the variance σ^2 is well-defined and governs the signal evolution. However, for $\alpha < 2$, the variance becomes infinite, rendering the traditional iterative map (Eq. 1.4) divergent. Lévy Mean Field Theory resolves this by parameterising the distribution of neural inputs h_i^l using the *scale parameter* q^l , leading to a new recursion relation:

$$q^l = D_w \int |\phi(z)|^\alpha p_{S_\alpha((q^{l-1}/2)^{1/\alpha})}(z) dz + D_b \quad (1.7)$$

In Eq. 1.7, D_w represents the coupling strength, acting as the functional successor to Gaussian variance. This map allows for the characterisation of fluctuations in heavy-tailed networks, providing a theoretical tool to predict signal stability in regimes where standard MFT fails.

1.3.2 The Emergence of Extended Criticality

The most striking prediction of Lévy Mean Field Theory is the existence of an *extended critical regime*. Unlike Gaussian networks, which possess a singular "edge-of-chaos" line requiring precise fine-tuning, heavy-tailed networks ($\alpha < 2$) exhibit a broad region of parameter space where information can propagate to theoretically infinite depths. This extended criticality is a robust feature of heavy-tailed architectures, allowing the network to maintain a balance between representation expansion and contraction without sensitive parameter selection.

Spectral analysis of the layer-wise Jacobians in this regime reveals that the eigenvectors are *spatially multifractal*. This implies a mixture of localised and delocalised properties, providing a microscopic mechanism for the network to separate crucial features from random noise. This multifractality allows for a "hierarchical division of labour" within the network, where different layers can perform distinct computational tasks based on their local transport physics.

1.3.3 The Gap: Non-Equilibrium Training Dynamics

Although Qu et al. (2022) successfully established the stationary properties of the extended critical regime, the *non-equilibrium dynamics* occurring during the actual training process remain largely unexplored. While the theory predicts stable signal propagation at initialisation, it does not account for how the weight manifold evolves, or how specific layers escape local minima through discrete *Lévy flights* during the learning trajectory.

The primary aim of this thesis is to fill this gap by empirically characterising the transport kinetics of heavy-tailed neural networks. By monitoring layer-wise displacement tracking and Time-Averaged Mean Squared Displacement (TAMSD), we seek to identify the dynamical signatures—such as ballistic transport and spatial decoupling—that enable the superior performance and stability of these architectures in traditionally "silent" ordered regimes.

Chapter 2: Methodology

Chapter 3: Results and Discussion

Chapter 4: Conclusion

Appendix A: Computer Code

Relevant snippets of the PyTorch implementation and the Lévy-stable initialization logic are included here. The full repository can be found at: <https://github.com/YiHaoFromEarth/ht-initialisation-dynamics>